

Yige Yan

Ph.D. Candidate, College of Computing and Data Science, Nanyang Technological University

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Research Profile

My research focuses on **pathology–age decoupling** for neurodegenerative diseases in brain functional connectivity and **contrast-free microvascular obstruction identification** from cine cardiac MRI. Seeking research internships (medical AI, multimodal learning, LLMs, AI agents, computer vision) before thesis submission in the first half of 2027, and postdoctoral positions from 2027.

Education

Nanyang Technological University Singapore

Ph.D. in Computer Science and Engineering, GPA: 4.67/5.00 Aug 2023 – Present

Main supervisor: Prof. Jagath C. Rajapakse (IEEE Fellow), College of Computing and Data Science.

Co-supervisor: Assoc. Prof. Nagaendran Kandiah, Lee Kong Chian School of Medicine.

Expected thesis submission: first half of 2027.

Nanyang Technological University Singapore

M.Sc. in Computer Control and Automation, GPA: 4.44/5.00 Aug 2022 – Jun 2023

Supervisor: Prof. Wee Peng Tay, School of Electrical and Electronic Engineering.

Hohai University Nanjing, China

B.Eng. in Computer Science, GPA: 4.48/5.00 Sep 2017 – Jun 2021

Supervisor: Dr. Xiaoqin Zhou, College of Artificial Intelligence and Automation.

Publications & Manuscripts

Peer-reviewed

1. **Yige Yan**, Yi Hao Chan, Jun Cheng, Qian Hui Chew, Kang Sim, and Jagath C. Rajapakse. “Latent Pathology–Age Decoupling in Functional Connectivity Graphs for Multi-task Schizophrenia Assessment.” *MICCAI*, 2026. **Early Accept.**
2. **Yige Yan**, Jun Cheng, Xulei Yang, Shuang Leng, Ru San Tan, Liang Zhong, and Jagath C. Rajapakse. “Dual Correlation-Aware Mamba for Microvascular Obstruction Identification in Non-contrast Cine Cardiac Magnetic Resonance.” *MICCAI*, 2025. **Early Accept.**
3. **Yige Yan**, Jun Cheng, Xulei Yang, Zaiwang Gu, Shuang Leng, Ru San Tan, Liang Zhong, and Jagath C. Rajapakse. “Coarse-Grained Mask Regularization for Microvascular Obstruction Identification from Non-contrast Cardiac Magnetic Resonance.” *MICCAI*, 2024. **Early Accept.**

Under review

4. **Yige Yan**, Jing Xia, Yi Hao Chan, Thomas Welton, and Jagath C. Rajapakse. “Decoupling Age Effects from Parkinson’s Disease Functional Connectivity.” *Medical Image Analysis*.
5. **Yige Yan**, Jun Cheng, Xulei Yang, Zaiwang Gu, Shuang Leng, Ru San Tan, Liang Zhong, and Jagath C. Rajapakse. “Spatiotemporal Adaptive Mamba with Coarse-Grained Mask Regularization for Microvascular Obstruction Identification from Non-contrast Cine Cardiac MRI.” *Neurocomputing*.

In preparation

6. **Yige Yan**, Jing Xia, Yi Hao Chan, Nagaendran Kandiah, and Jagath C. Rajapakse, for the Alzheimer’s Disease Neuroimaging Initiative. “Clinical Evidence Reconfigures Age Reliance in Language Models Assessing Alzheimer’s Disease.” *Target: npj Digital Medicine*.
7. **Yige Yan**, Pei Yang, Jun Cheng, Lianggeng Gong, and Jagath C. Rajapakse. “Temporal Variation Proxy and Staged Proposal–Support Integration for Contrast-Free MVO Segmentation in Cine CMR.” *Target: IEEE Transactions on Medical Imaging*.
8. **Yige Yan**, Hang Wang, Yi Hao Chan, and Jagath C. Rajapakse. “Auditing Named Knowledge Interfaces: Regularization, Responsiveness, and Predictive Utility.” *Target: Knowledge-Based Systems*.

Research Experience

Nanyang Technological University

Singapore

Ph.D. Researcher, College of Computing and Data Science

Aug 2023 – Present

- **Contrast-free cardiac MRI.** Built a multi-year research line on recovering microvascular obstruction (MVO) information from cine MRI alone, starting from coarse-grained supervision under imperfect LGE-to-cine registration and progressing through motion modeling and spatiotemporal state-space learning to pixel-level MVO segmentation.
- **Confounder-free functional connectivity analysis.** Disease-related signal is present but often dominated by age-related variation. Designed graph-based representation learning methods that decouple disease effects from age effects and further reintroduce age into the model as controlled clinical context.

Institute for Infocomm Research (I²R), A*STAR

Singapore

Research Student Attachment, Medical Imaging

Aug 2023 – Present

Mentor: Dr. Jun Cheng.

- Microvascular obstruction identification from non-contrast cine cardiac MRI, with collaborators across A*STAR, NTU, Duke-NUS Medical School and the National Heart Centre Singapore.

Institute for Infocomm Research (I²R), A*STAR

Singapore

Research Student Attachment, OCTA Denoising

Mar 2023 – Jun 2023

- Deep-learning denoising and enhancement for optical coherence tomography angiography (OCTA), with collaborators across A*STAR, NTU and the Singapore National Eye Centre.

Research Grants & Proposal Experience

Singapore MOE Academic Research Fund (AcRF) Tier 2

In preparation, Sep 2026

Primary proposal writer and research-plan developer

Faculty PI: Prof. Jagath C. Rajapakse

“Separating and Contextualizing Age for Reliable Assessment and Prognosis of Neurodegenerative Disease.”
Drafted the research plan and preliminary-study design, extending my doctoral research on pathology–age decoupling into a multi-project research program.

Conference Presentations & Posters

- *MICCAI 2026*, Strasbourg, France, Sep–Oct 2026. Poster presentation.
- *MICCAI 2025*, Daejeon, Republic of Korea, Sep 2025. Poster presentation.
- *IEEE EMBS – Duke-NUS – HealthTEC.SG – NUS iHealthtech Forum 2025: AI in Healthcare and MedTech*, Singapore, Apr 2025. Poster presentation.
- *MICCAI 2024*, Marrakesh, Morocco, Oct 2024. Poster presentation.
- *Singapore Healthcare Engineering Conference (SHEC) 2024*, IEEE EMB Singapore Chapter, Singapore, Oct 2024. Poster presentation.
- *AI Health Summit 2023*, SingHealth, Ministry of Health and A*STAR, Singapore, Nov 2023. Poster presentation.

Teaching Experience

Nanyang Technological University

Singapore

Teaching Assistant, College of Computing and Data Science

2024 – Present

- SC4001 Neural Networks and Deep Learning
- SC1013 Physics for Computing
- SC1006 Computer Organisation & Architecture
- SC2079 Multidisciplinary Design Project
- SC2008 Computer Network

Research Mentoring

Nanyang Technological University

Mentor, Undergraduate Final-Year Projects (FYP)

Singapore

2025 – Present

- AY2026/27. “Predicting Multiple Cognitive Functions Simultaneously from Brain Functional Connectivity via an Interpretable Multi-Task Learning Network.”
- AY2026/27. Age-conditioned Riemannian residual modeling of schizophrenia functional connectomes.
- AY2025/26. “Explainable AI for Functional MRI Deep Learning Model Analysis.”
- AY2025/26. “Multi-Modal Functional MRI Fusion for Fluid Intelligence Prediction.”
- AY2025/26. Graph state-space modeling for Alzheimer’s disease neuroimaging.

Honors, Awards & Scholarships

- **MICCAI 2024 Registration and Travel Award.**
- **Nanyang Technological University Research Scholarship (RSS).**
- **Outstanding Undergraduate Graduation Project**, Hohai University, 2021.
- **Outstanding Student Award**, Hohai University.
- **Academic Merit Scholarship**, Hohai University.

Technical Skills

Programming & Frameworks: Python, PyTorch, PyTorch Geometric, C++, C, NumPy, SciPy, scikit-learn.

Methods: Deep learning, graph neural networks, state-space models (Mamba), medical image analysis, multimodal learning, statistical evaluation.

Domains: Medical image analysis, cardiac MRI, fMRI, sMRI, neuroimaging, OCTA, medical multimodal AI.

Languages: Chinese (native), English (fluent), Japanese (proficient).